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Antibiotic Exposure and Reduced Short Chain Fatty Acid Production after Hematopoietic Stem Cell Transplant



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ABSTRACT

Human studies have shown loss of diversity of the gut microbiome following hematopoietic stem cell transplantation (HSCT) in association with significant gut injury caused by the preparative regimen. Prolonged antibiotic use worsens loss of microbiome diversity and increases risk of complications such as graft-versus-host disease (GVHD). Our data support the hypothesis that loss of intestinal commensals that produce short-chain fatty acids (SCFAs) may increase dysbiosis. Here, we report an extensive longitudinal examination of changes in the luminal SCFAs in children undergoing allogeneic HSCT, and the relationship of those changes to the microbiota and antibiotic exposure. We found significant and progressive alterations in butyrate, and in additional SCFAs in stool in the first 14 days after transplant, a finding not observed in published mouse studies. SCFA levels were lower in children receiving antibiotics with activity against anaerobic organisms. Moreover, day 14 post-HSCT butyrate and propionate levels are lower in children who went on to develop GVHD, although our disease population was small. These data provide insight into the mechanism of prior observations that loss of diversity and increased antibiotic use are associated with GVHD following HSCT. Our findings offer potential modifiable targets to reduce risk of GVHD and improve survival after HSCT.

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INTRODUCTION

Disruption of the gut microbiome has been associated with several disease pathologies, including inflammatory bowel disease, autism, and obesity [1–3]. Similarly, loss of diversity of the microbiome had been associated with increased mortality after allogeneic hematopoietic stem cell transplantation (HSCT). Moreover, the gut microbiome is disturbed in gut graft-versus-host disease (GVHD), a potentially lethal complication of HSCT [4,5].

Gut injury is essential to many of the toxicities of HSCT, and restoration of intestinal homeostasis is essential for resolution of tissue injury, establishment of immunological tolerance, and patient survival [6]. Studies have shown that the intestinal microbiota composition and diversity is substantially altered, with loss of anaerobes being a significant finding [7–9]. The

close relationship among the intestinal epithelium, the luminal microbiota, and innate and adaptive immunity make a causative role for microbiome changes in the presence or absence of immunological tolerance after HSCT plausible [10]. Short-chain fatty acids (SCFAs) are known to play an important role in the interaction between the microbiota and host immune cells in the resting state [11–13]. SCFAs are not made by humans and are only derived from the gut microbiota. Some SCFAs (particularly butyrate) are a preferred energy source for intestinal epithelium cells, and butyrate plays a key role in the maintenance of physiological hypoxia in the epithelial border of the intestine [14,15]. Studies in other colitides have shown altered SCFA production and support a direct role in the exacerbation of disease [16].

Reddy et al. [17] have recently reported on the role of SCFAs in allogeneic and syngeneic transplant in an important study performed in mice. Perhaps surprisingly, this study reports no change in luminal SCFA levels between mice receiving allogeneic and syngeneic transplants. However, levels of butyrate were reduced in the intestinal wall of mice with GVHD due to

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reduced expression of an SCFA transporter and SCFA receptor molecule in the intestine of the mice. We hypothesize that, in contrast to recent mouse data, the human gut microbiome and gut-derived SCFAs will be significantly altered following transplant, and that this perturbation may be further exacerbated in those individuals who go on to have adverse outcomes, such as GVHD. Here, we report our analysis of SCFA in the stool of children undergoing allogeneic HSCT in the first weeks after transplant, and explore the role of antibiotic exposure.

METHODS

Human Subjects

All biological samples from study participants (N = 42) were obtained through a Cincinnati Children's Hospital Medical Center Institutional Review Board approved study under informed written consent. Samples were collected weekly after admission for HSCT; exact day of stool sample collection was necessarily dependent on stool production. Clinical data were collected prospectively and retrieved from the HSCT database, including patient demographics, disease and therapy characteristics, and transplant complications. Patients were consented before the start of the preparative regimen. Samples were obtained at baseline (before preparative regimen), day 0 (day of transplant), day 7 post-transplant, and day 14 post-transplant. Currently accepted clinical criteria were used for diagnosis of acute GVHD [18], overall survival, and transplant-related mortality [19]. The patients included in the study were consecutive cases with adequate available stool samples to allow for both SCFA quantification and microbiome analysis. No restrictions or exclusions of patient samples were made based on conditioning regimen, disease type, GVHD prophylaxis, graft source, or organ affected. Prophylactic antibiotics were not prescribed. Broad-spectrum antibiotics (typically first-line piperacillin or tazobactam, second-line meropenem) were administered with first fever and continued until count recovery. Aminoglycosides and anaerobic antibiotics such as metronidazole and clindamycin were administered if clinically indicated. Antibiotic exposures for each patient were calculated as follows: a dose of an antibiotic agent of any quantity during one hospital day was recorded as 1 day of exposure to that specific drug. The days of exposure to any drug with activity against anaerobes was totaled and is expressed as days of anaerobic antibiotic exposure.

Fecal Sample Preparation for Nuclear Magnetic Resonance

Fecal test samples were aliquoted and weighted into 2-mL bead tubes containing 2.8-mm ceramic beads (VWR, Radnor, PA) before the sample processing. On the day of processing, 1 mL of cold PBS was added to the sample, homogenized for 30 seconds at 5000 rpm using Minilys (Bertin Instruments, Montigny-le Bretonneux, France). The samples were then transferred into a 2-mL tube and centrifuged at $10,000 \times g_n$ at 4°C for 10 minutes. Supernatants (400 μ L) were then filtered at $12,000 \times g_n$ for 60 minutes at 4°C using pre-washed 3-kDa spin filters (NANOSEP 3K, Pall Life Sciences, Portsmouth, United Kingdom). A buffer containing 100 mM phosphate buffer in D₂O (deuterium oxide), pH 7.3, and 1.0-mM TMSP (3-trimethylsilyl 2,2,3,3-d₄ propionate) was added to 180 μ L of the filtered feces. The final sample volume was 200 μ L and the final TMSP concentration in each sample was .1 mM.

Nuclear Magnetic Resonance Spectroscopy Data Acquisition and Processing

The experiments were conducted using 180- μ L samples in 103.5 mm $\times$ 3 mm nuclear magnetic resonance (NMR) tubes (Bruker Corporation, Billerica, MA). One-dimensional ¹H NMR spectra were acquired on a Bruker Avance II 600 MHz spectrometer. All data were collected at a calibrated temperature of 298 K using a 3-pulse sequence based on the noesypr1d (or noesygppr1d) pulse sequence in the Bruker pulse sequence library. This pulse sequence provided water suppression with good baseline characteristics. Experiments were run with 4 dummy scans and 128 acquisition scans, with an acquisition time of 3.4 s and a relaxation delay of 3.0 s for a total repetition cycle (acquisition time + relaxation delay) of 6.4 s. The NOESY mixing time was 6 ms. The spectral width was 16 ppm, and 64,000 real data points were collected. Under automation control, each sample analysis took about 10 minutes for setup and 30 minutes for acquisition. During the 10-minute setup time, the temperature was monitored for equilibration. All standards for free induction decay (FIDs) were subjected to an exponential line broadening of .3 Hz. Upon Fourier transformation, each spectrum was manually phased, baseline corrected, and referenced to the internal standard TMSP at .0 ppm for polar samples using Topspin 3.5 software (Bruker Analytik, Rheinstetten, Germany).

For a representative sample, 2-dimensional data were collected for metabolites assignment. ¹H-¹H total correlation spectroscopy data, a relaxation delay equal to 2 s, isotropic mixing time of 80 ms with a B₁ field strength of 10 kHz were used for 2048 data points with 128 scans per increment were acquired with spectral widths of 14 ppm. Also, 2D ¹H-¹³C heteronuclear single quantum coherence (HSQC) data, a relaxation delay equal to 1.5 s was used between acquisitions and a refocusing delay of 3.45 ms was

implemented. The 2048 data points with 128 scans per increment were acquired with spectral widths of 11 ppm in F2 and 180 ppm in F1 (¹³C).

Chemical shifts were assigned to metabolites based on 1-dimensional ¹H and 2-dimensional total correlation spectroscopy, and HSQC NMR experiments with reference spectra found in databases; Human Metabolome Database [20]; and Chenomx NMR Suite profiling software (version 8.1, Chenomx, Edmonton, Canada). The concentrations of the metabolites were calculated using Chenomx software based on the internal standard (TMSP). All metabolite concentrations were normalized to the dry fecal weight of the sample. We assigned and calculated the concentrations of SCFAs known to be derived from gut microbiota rather than the human host [21] before HSCT (baseline), on the day of transplant (day 0), 7 days after HSCT (day 7), and 14 days post-HSCT (day 14).

16S. Deep Sequencing and Microbial Analysis

On indicated days, fecal samples aliquots were collected from patients and stored at -80°C. DNA was extracted and purified using the QiaAmp Power Fecal Kit (Qiagen, Hilden, Germany). 16S sequencing was performed by the DNA Core at Cincinnati Children's Hospital Medical Center. In short, the 319F-806R primer set was sequenced as a paired-end 300 run (PE300) on an Illumina MiSeq (Illumina, San Diego, CA). PCR was run with untailored primers (319F-ACTCTACGGGAGGCAGCAG and 806R-GGACTACHVGGGTWTC-TAAT) as a quality control for the 16S metagenomics DNA. The next-generation sequencing DNA library was prepared by PCR and loaded onto the MiSeq for sequencing. Data generated were presented in fastq format. The analysis was performed as described [8, 22, 23]. Comparisons between taxa were performed at the taxonomic level of Family or Order and were designated as butyrate producers if the major constituents were known to produce butyrate from previously published literature [24–26]. Major butyrate-producing bacteria fall within the Bifidobacteriaceae, Bacteroidaceae, Porphyromonadaceae, Eubacteriaceae, Lachnospiraceae, and Ruminococcaceae, while Erysipelotrichaceae and Fusobacteriaceae have also been reported to produce butyrate.

RNA Sequencing and Analysis of Colonic Tissue

Colonic biopsies were collected with informed consent at the time of clinically indicated endoscopic procedures in children recovering from transplant. RNA was extracted from gastrointestinal mucosal tissue using RNeasy Mini Kit (Qiagen Sciences, Germantown, MD) according to the manufacturer's protocol. RNA sequencing analysis was performed in the University of Cincinnati Department of Environmental Health Core facility essentially as described previously [27]. Briefly, samples were sequenced using HiSeq1000 (Illumina) to obtain 50-bp single-end reads. Quality control steps were included to determine total RNA quality and quantity, cDNA fragment size, and optimal number of PCR preamplification cycles. Samples that failed quality control were eliminated from further downstream steps. RNA sequencing data were mapped against the hg19 reference using TopHat [28]. Gene expression profiles were compared between 2 children with GVHD and 2 without GVHD. All *P* values presented are adjusted for false discovery.

Statistics

Categorical and continuous data are summarized by frequency (percentage) and mean $\pm$ SE. Univariate comparisons were compared between variables with 2 groups with *t*-tests or Wilcoxon rank sum test if the data contained significant outliers. Analysis of variance was used for comparisons with more than 2 groups. Comparisons for variables taken at multiple time points were compared with baseline using paired *t*-tests. With respect to Figure 3, subcategorization of patients into high and low luminal butyrate groups was defined as having >1 mM butyrate/dry fecal weight or <1 mM butyrate/dry fecal weight. This subcategorization was made to allow for near equal distribution of the 2 groups at day 14. Utilization of the value of 1-mM butyrate/dry fecal weight resulted in 48% of patients in the high-butyrate group and 52% of patients in the low-butyrate group. All metabolite concentrations were normalized to dry tissue weight and log transformed before analyses unless otherwise indicated. Statistical analysis performed for Figure 5 was not log transformed, as a result of small sample numbers and the presence of a high number of "zero" values. An unpaired, 2-tailed *t*-test with Welch's correction was used for all comparisons of patients who developed GVHD with those who did not. All graphs were generated using GraphPad Prism (GraphPad Software Version 7, La Jolla, CA).

Data Availability

The RNA sequencing data discussed (Table 2) have been deposited in the National Center for Biotechnology Information's Gene Expression Omnibus [29, 30] and are accessible through Gene Expression Omnibus Series accession number GSE102228.

RESULTS

Reduction in Microbial Metabolites following HSCT

The demographics of the study population are presented in Table 1. We analyzed the luminal contents (stool) of the intestines with NMR spectroscopy. Fecal butyrate levels were significantly decreased at both day 7 and day 14 following HSCT compared with baseline fecal SCFA levels (Figure 1A). Similarly, significant reductions in fecal propionate levels were observed in the days following transplant (Figure 1B). Acetate levels in the stool were also reduced in the days following HSCT (Supplementary Figure S1A). Taken together, these data show that key microbial metabolites (acetate, butyrate, and propionate) are commonly reduced in luminal contents following allogeneic HSCT.

Evaluation of Risk Factors for Reduced SCFA Levels following HSCT

Univariate analyses were performed to evaluate the impact of a number of variables (age, reduced-intensity conditioning versus myeloablative conditioning, and antibiotic exposure) on microbial metabolite levels. We found no significant correlations between reduced butyrate levels at day 14 post-HSCT and the age of the patient or the intensity of conditioning. Previous studies have demonstrated a significant impact of antimicrobial agents on gut microbiota and microbial metabolites and end products [31–33]. To corroborate and perhaps better explain the mechanism of previous findings, we categorized patients into 2 groups (high and low luminal butyrate or propionate) based on the metabolomics findings of the stool samples. We then examined the difference between the total anaerobic antibiotic exposures of the patients and the microbial metabolite levels in the stool samples. We found a significant association between low butyrate (Figure 2A) and low propionate (Figure 2B) levels at day 14 post-HSCT and higher total exposure to antibiotics that target anaerobic bacteria. Luminal acetate levels were also lower with higher total exposure to antibiotics, although not to a significant degree (Supplementary Figure S2A).

Correlations between Microbiome and Butyrate Levels at Day 14 after HSCT

We next used metagenomic analysis of the luminal contents to correlate SCFA levels with the microbial environment. As expected, the diversity of the gut microbiota was significantly decreased following HSCT (Figure 3A). We categorized the observed bacteria in our samples into 2 groups (high and low butyrate producers) and then examined the difference between the relative representation of each category in a stool sample and butyrate level. We found that the group with higher luminal butyrate also had greater representation of butyrate producers in the microbiome (Figure 3B).

Elevated Microbial and Dietary Metabolites following HSCT

In addition to identifying a decrease in SCFAs after HSCT, we also observed a significant increase in formate levels post-HSCT (Figure 4A). Moreover, we found a significant correlation between those individuals with higher formate levels at day 14 and higher levels of a family of bacteria within the Proteobacteria phylum *Enterobacteriaceae* (Figure 4B).

We also found that fecal and urine tryptophan levels were elevated in the days following HSCT (Figure 4C, D). Luminal tryptophan is metabolized to gut-derived serotonin that has important functional effects on gastrointestinal function, particularly motility [34–36].

Table 1

Demographics of the Study Population

	Patients (n = 42)
Male	27 (64)
Age, yr	7.1 (.8–27.9)
Diagnosis	
• Bone marrow failure	13 (31)
• Malignancy	12 (29)
• Immune deficiency	8 (19)
• Benign hematology	8 (19)
• Genetic/metabolic	1 (2)
Conditioning regimen	
• Reduced-intensity conditioning	14 (33)
• Myeloablative conditioning	28 (67)
Stem cell source	
• Bone marrow	28 (67)
• Cord	1 (2)
• PBSC	13 (31)
Donor	
Autologous	6/42 (14)
Allogeneic	36/42 (86)
• Matched related donor	12/36 (33)
• Matched unrelated donor	13/36 (36)
• Mismatched unrelated donor	11/36 (31)
GVHD prophylaxis	
• CSA/TAC + prednisone	10 (28)
• CSA/TAC + MTX	8 (22)
• Ex vivo T cell depletion	6 (17)
• CSA/TAC + MTX + abatacept	4 (11)
• CSA/TAC + MMF	4 (11)
• CSA/TAC + prednisone + ATG	3 (8)
• CSA/TAC + MMF + maraviroc	1 (3)
Day 100 GVHD	
• Grade I	5 (12)
• Grade II	3 (7)
• Grade III–IV	2 (5)
Day 100 GVHD organ involvement	
• Skin	9 (21)
• Gut	2 (5)
• Liver	0 (0)

Values are n (%), median (range), or n/n (%).

PBSC indicates peripheral blood stem cell; CSA, cyclosporine; TAC, tacrolimus; MTX, methotrexate; MMF, mycophenolate mofetil; ATG, antithymocyte globulin.

SCFAs and GVHD

Of the 42 patients participating in the study, 10 (24%) went on to develop GVHD. The concentrations of butyrate (median: control 1.77 mM/g dry weight, GVHD .0550 mM/g dry weight) (Figure 5A), propionate (median: control 6.63 mM/g dry weight, GVHD .208 mM/g dry weight) (Figure 5B), and acetate (median: control 39.6 mM/g dry weight, GVHD 7.92 mM/g dry weight) (Figure 5C) in the stool at day 14 after HSCT were significantly reduced in individuals that went on to later develop GVHD at any site. Formate levels (median: control .290 mM/g dry weight, GVHD .433 mM/g dry weight) in the stool at day 14 post-HSCT were not significantly altered in those patients who developed GVHD compared with those who did not (Figure 5D).

Expression of Microbial Metabolite Transporters in Mucosal Biopsies from Children with GVHD

Next, we compared RNA expression levels of butyrate transporters and receptors in colonic biopsies from pediatric transplant recipients who either did or did not develop GVHD. We found that expression of SLC5A8 was unchanged in the gastrointestinal mucosa of children with GVHD in our study, but found a significant increase in expression of SLC16A3 and GPR43 (Table 2). Expression of SLC5A12 was significantly reduced in the colon of children with GVHD. We saw a modest

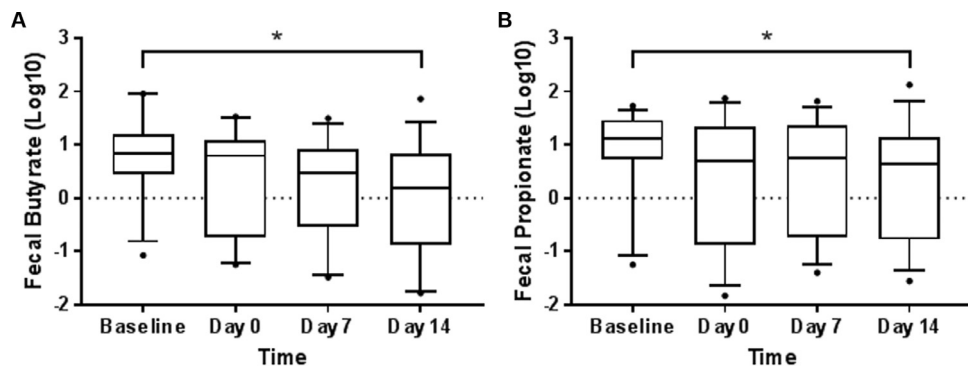


Figure 1. Reduction in microbial metabolites following HSCT. Levels of the SCFAs butyrate (A) and propionate (B) in stool in the first 14 days after HSCT are shown on a log scale. The level of butyrate is significantly lower at day 14 compared with the baseline sample collected before start of HSCT ($P = .0039$). Similarly, the level of propionate is significantly lower at day 14 compared with the baseline sample collected before start of HSCT ($P = .0086$). Points within graphical plots represent individual patient data and bars represent the 5th to 95th percentiles.

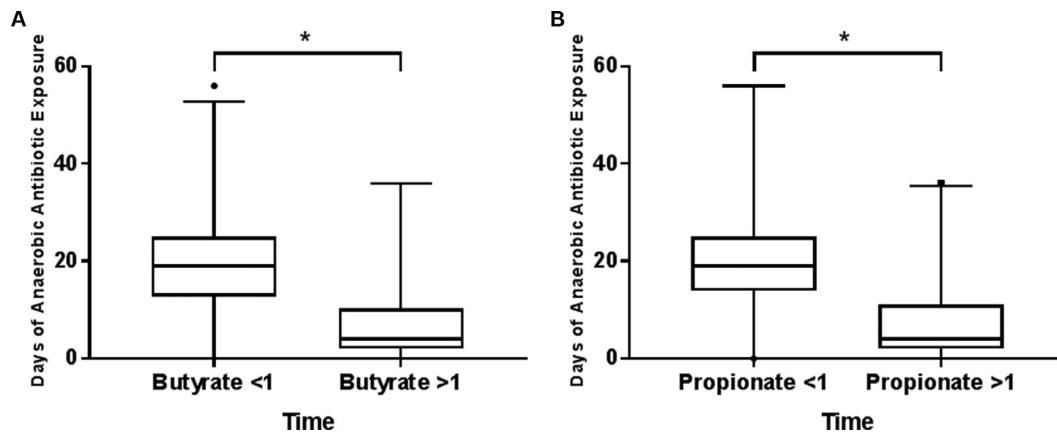


Figure 2. Significant correlations between anaerobic antibiotic exposure and levels of butyrate and propionate at day 14 post-HSCT. The difference between the total anaerobic antibiotic exposures of the patients and the microbial metabolite levels in the stool samples was examined to determine the impact that these drugs have on microbial metabolite composition. (A) Significantly lower levels of butyrate (butyrate <1, $P = .0025$) are associated with a greater number of days of anaerobic antibiotic exposure. (B) Significantly lower levels of propionate (propionate <1, $P = .0012$) are also linked to a greater number of days of anaerobic antibiotic exposure.

decrease in expression of ACMS3 and GPR41, and no change in expression of SLC16A1, ACADS, ECHS1, HSD17B10, ACAT2, and GPR109A.

DISCUSSION

The mechanism by which changes in the microbiota contribute to intestinal inflammation and mortality has been unproven in humans. A reasonable hypothesis that has been proffered is that loss of intestinal Firmicutes reduces intestinal butyrate, leading to loss of physiological hypoxia in the colon. Loss of these obligate anaerobes allows expansion of facultative anaerobes, in particular, Enterobacteriaceae and Lactobacillaeae, and promotion of local inflammation in the gut and translocation of bacterial molecules into the circulation that further aggravate inflammation and adverse outcomes, such as GVHD. In contrast to this hypothesis, a recent murine study showed that there was no change in butyrate in the intestinal lumen contents in mice with and without GVHD. However, butyrate was reduced in the intestinal tissue due to reduced expression of a butyrate transporter and a butyrate receptor [17]. Here, we report a study in children undergoing HSCT to try to directly address reduction of luminal SCFAs as a potential mechanism linking changes in the microbiota to post-transplant outcomes in humans. We concentrated our efforts on events in the early transplant period when clinical gut injury is most prominent, as we believe, as reported by others, that

translocation of bacterial products and danger-associated molecular patterns initiate acute GVHD in the gut, skin, and liver and contribute importantly to mortality after HSCT [24,37,38].

Our human data show important differences from the mouse study. We identified a progressive decline in SCFAs in the intestinal lumen in the first weeks after HSCT, with significant reduction in butyrate and propionate. Moreover, we examined the expression of a number of butyrate transporters and receptors in mucosal biopsies from children with GVHD and found no change in expression of most, and indeed increased expression of some, again in contrast to the mouse study. In addition, we saw a correlation between depletion of Firmicutes and other anaerobes, in particular, *Ruminococcus*, and lower levels of SCFAs. Importantly, we demonstrated an association between higher exposure to antibiotics with activity against anaerobic bacteria and lower butyrate and propionate, suggesting that reduced use of antianaerobic antibiotics after HSCT, when not essential, will improve outcomes. The striking differences between the mouse and human studies may reflect the controlled environment of the mouse studies, with inbred animals living in a pathogen-controlled environment eating a uniform diet.

In addition to depletion of butyrate and propionate, we also saw increased levels of formate in the weeks after HSCT. A

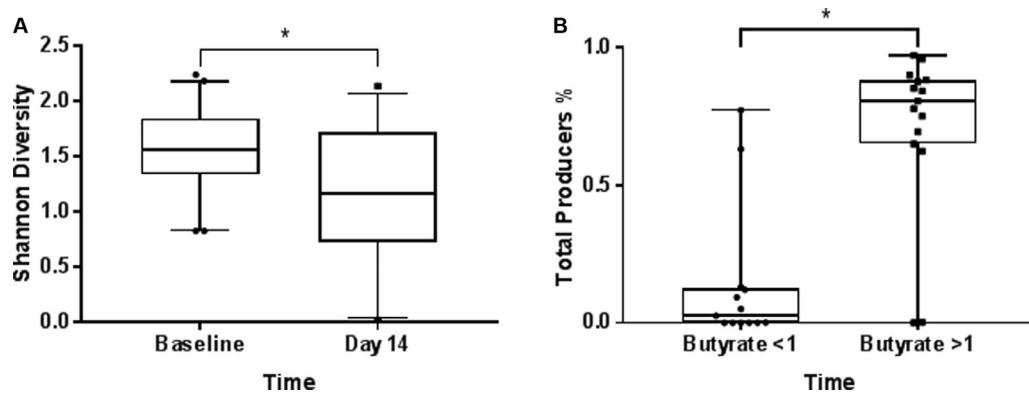


Figure 3. Significant correlations between microbiome and butyrate levels at day 14 post-HSCT. 16S deep sequencing technology was used to evaluate microbial diversity and determine correlations between microbial populations and butyrate. (A) Reduction in microbial diversity at day 14 after HSCT ($P = .0028$). (B) Higher and lower levels of butyrate are associated with higher and lower levels of butyrate-producing organisms ($P < .0001$), linking levels of SCFAs with changes in the microbiome. Points within graphical plots represent individual patient data and bars represent the 9th to 95th percentiles.

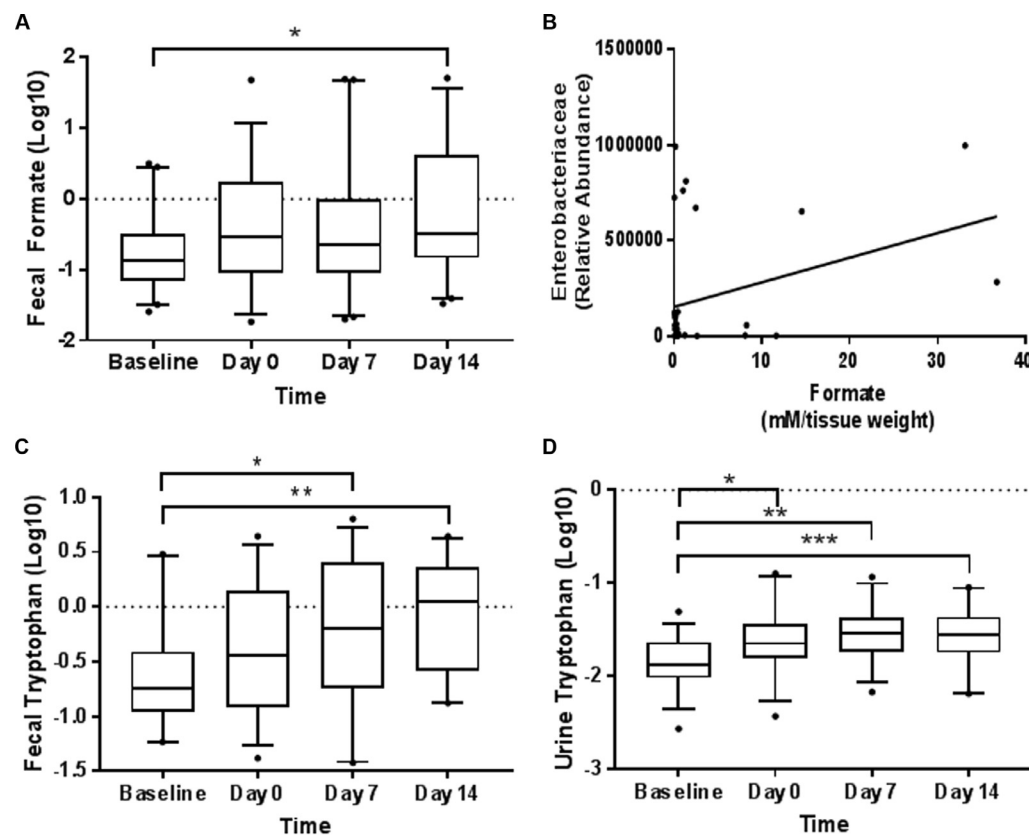


Figure 4. Elevated microbial and dietary metabolites, formate and tryptophan, following HSCT. (A) Progressive increase in luminal formate 14 days after HSCT ($P = .0034$). (B) A significant correlation between individuals with higher luminal formate levels and elevated proteobacteria in the gut microbiome ($P = .0442$). (C) Luminal tryptophan levels were significantly increased at both day 7 ($P = .0363$) and day 14 ($P = .0069$) when compared with baseline levels. (D) Urinary tryptophan levels also increased significantly at day 0 ($P = .0021$), and following HSCT at day 7 ($P < .0001$) and day 14 ($P = .0001$) when compared with baseline levels. Points within graphical plots represent individual patient data and the bars represent the 9th to 95th percentiles.

previous murine study of chemically induced colitis has shown increased formate levels in dysbiosis, and that the formate was microbe derived and contributed to the bloom of *Enterobacteriaceae* in the inflamed gut [39]. In agreement with this, we found a correlation between formate levels and presence of *Enterobacteriaceae* in the human gut. Little is known about the potential health effects of formate alterations. However, these findings suggest that modulation of formate levels and the bacterial species that produce this fatty acid may be important in HSCT patients.

We hypothesized that urinary biomarkers might reflect changes in the intestinal microbiota, so we performed NMR analysis of urine in parallel with our stool studies. We found a significant increase in urinary tryptophan in the weeks after HSCT, paralleled by increased tryptophan in the stool. Tryptophan is known to play an important role in intestinal homeostasis, as it is metabolized to gut-derived serotonin, which regulates gut motility and function [34–36,40]. Work conducted in humans with human immunodeficiency virus linked tryptophan catabolism to dysbiosis, expansion of intestinal

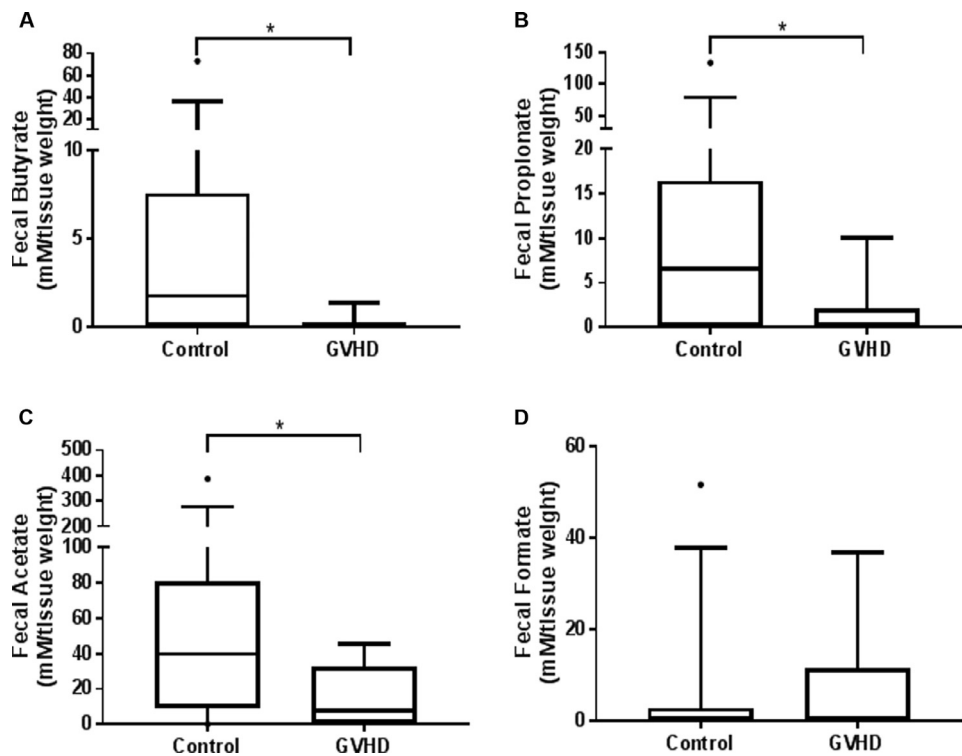


Figure 5. SCFAs are altered at day 14 post-transplant in patients who go on to develop GVHD. Levels of the SCFAs butyrate ($P = .0142$) (A), propionate ($P = .0108$) (B), and acetate ($P = .047$) (C) are significantly reduced at day 14 post-transplant in children who go on to develop GVHD when compared with children who do not develop GVHD. (D) Formate levels are slightly elevated in children with GVHD, but not to a significant degree. Points within graphical plots represent individual patient data and bars represent the 9th to 95th percentiles.

Table 2

Microbial metabolite transporters and receptors are altered in individuals that develop GVHD

Transporter	Expression	P Value
SLC5A8	Not significantly changed	.93
SLC16A3	Increased 3-fold	7.3e-5
SLC5A12	Decreased .3-fold	.008
SLC16A1	Not significantly changed	.89
ACMS3	Decreased .37-fold	.014
ECHS1	Not significantly changed	.89
HSD17B10	Not significantly changed	.73
ACAT2	Not significantly changed	.55
Receptors		
GPR43-FFAR2	Increased 4.5-fold	.013
GPR41-FFAR3	Decreased .12-fold	.007
GPR109A-HCAR2	Not significantly changed	.2

RNA sequencing analysis of expression was performed on RNA isolated from mucosal biopsies from 2 children with GVHD and 2 without GVHD. Expression of genes in the children with GVHD is shown relative to expression in children without GVHD.

proteobacteria, and translocation of bacterial molecules that maintain a state of chronic inflammation [41].

Mathewson et al. [37] reported decreased expression of a single butyrate transporter and a single butyrate receptor as a possible explanation for the reduced butyrate they saw in colonic tissue from mice with GVHD. We examined expression of 8 different butyrate transporters and 3 receptors. We found no change in expression of the specific transporter and receptor noted to be reduced in the Mathewson study, and variable changes in the expression of other transporters and receptors, with increases and decreases in expression seen. The importance, if any, of alterations in butyrate transport in the human colon in GVHD is unclear from our study.

In summary, our data provide insights into a potential mechanism for the observation that microbiota changes after HSCT may lead to increased risk of gut GVHD and mortality. Due to an overall lower than average incidence of GVHD in our study population, likely due to the young age of the children, these findings warrant further exploration in additional datasets to further explore the negative impacts that gut microbial perturbations and loss of SCFAs have in patient outcomes. Our data support clinical trials to maintain the diversity of the microbiota, perhaps with enteral supplements and reduce the loss of anaerobes. Moreover, antibiotics should be selected judiciously and used sparingly to maintain the microbiota after HSCT.

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SUPPLEMENTARY DATA

Supplementary data related to this article can be found online at doi:10.1016/j.bbmt.2018.07.030.

REFERENCES

- Nishida A, Inoue R, Inatomi O, Bamba S, Naito Y, Andoh A. Gut microbiota in the pathogenesis of inflammatory bowel disease. *Clin J Gastroenterol*. 2018;11(1):1–10.
- Campion D, Ponzo P, Alessandria C, Saracco GM, Balzola F. Role of microbiota in the autism spectrum disorders [e-pub ahead of print]. *Minerva Gastroenterol Dietol*. 2018. <http://dx.doi.org/10.23736/S1121-421X.18.02493-5>. accessed August 15, 2018.
- Tagliabue A, Elli M. The role of gut microbiota in human obesity: recent findings and future perspectives. *Nutr Metab Cardiovasc Dis*. 2013;23(3):160–168.
- Shallis RM, Terry CM, Lim SH. Changes in intestinal microbiota and their effects on allogeneic stem cell transplantation. *Am J Hematol*. 2018;93(1):122–128.
- Shono Y. Gut microbiota and graft-versus-host disease: broad-spectrum antibiotic use increases post-allogeneic hematopoietic stem cell transplant graft-versus-host disease-related mortality. *Rinsho Ketsueki*. 2017;58(7):835–842.
- Hill GR, Ferrara JL. The primacy of the gastrointestinal tract as a target organ of acute graft-versus-host disease: rationale for the use of cytokine shields in allogeneic bone marrow transplantation. *Blood*. 2000;95(9):2754–2759.
- Jenq RR, Ubeda C, Taur Y, et al. Regulation of intestinal inflammation by microbiota following allogeneic bone marrow transplantation. *J Exp Med*. 2012;209(5):903–911.
- Taur Y, Jenq RR, Perales MA, et al. The effects of intestinal tract bacterial diversity on mortality following allogeneic hematopoietic stem cell transplantation. *Blood*. 2014;124(7):1174–1182.
- Simms-Waldrip TR, Sunkersett G, Coughlin LA, et al. Antibiotic-induced depletion of anti-inflammatory Clostridia is associated with the development of graft-versus-host disease in pediatric stem cell transplantation patients. *Biol Blood Marrow Transplant*. 2017;23(5):820–829.
- Sheehan D, Shanahan F. The gut microbiota in inflammatory bowel disease. *Gastroenterol Clin North Am*. 2017;46(1):143–154.
- Kelly CJ, Zheng L, Campbell EL, et al. Crosstalk between microbiota-derived short-chain fatty acids and intestinal epithelial HIF augments tissue barrier function. *Cell Host Microbe*. 2015;17(5):662–671.
- Nastasi C, Candela M, Bonefeld CM, et al. The effect of short-chain fatty acids on human monocyte-derived dendritic cells. *Sci Rep*. 2015;5:16148.
- Chang PV, Hao L, Offermanns S, Medzhitov R. The microbial metabolite butyrate regulates intestinal macrophage function via histone deacetylase inhibition. *Proc Natl Acad Sci U S A*. 2014;111(6):2247–2252.
- Rivera-Chavez F, Zhang LF, Faber F, et al. Depletion of butyrate-producing Clostridia from the gut microbiota drives an aerobic luminal expansion of Salmonella. *Cell Host Microbe*. 2016;19(4):443–454.
- Espey MG. Role of oxygen gradients in shaping redox relationships between the human intestine and its microbiota. *Free Rad Biol Med*. 2013;55:130–140.
- Sun M, Wu W, Liu Z, Cong Y. Microbiota metabolite short chain fatty acids, GPCR, and inflammatory bowel diseases. *J Gastroenterol*. 2017;52(1):1–8.
- Mathewson ND, Jenq R, Mathew AV, et al. Gut microbiome-derived metabolites modulate intestinal epithelial cell damage and mitigate graft-versus-host disease. *Nat Immunol*. 2016;17(5):505–513.
- Glucksberg H, Storb R, Fefer A, et al. Clinical manifestations of graft-versus-host disease in human recipients of marrow from HL-A-matched sibling donors. *Transplantation*. 1974;18(4):295–304.
- Smith AR, Majhail NS, MacMillan ML, et al. Hematopoietic cell transplantation comorbidity index predicts transplantation outcomes in pediatric patients. *Blood*. 2011;117(9):2728–2734.
- Wishart DS, Tzur D, Knox C, et al. HMDB: the Human Metabolome Database. *Nucleic Acids Res*. 2007;35:D521–D526.
- Ganapathy V, Thangaraju M, Prasad PD, Martin PM, Singh N. Transporters and receptors for short-chain fatty acids as the molecular link between colonic bacteria and the host. *Curr Opin Pharmacol*. 2013;13(6):869–874.
- DeSantis TZ, Hugenholtz P, Larsen N, et al. Greengenes, a chimera-checked 16S rRNA gene database and workbench compatible with ARB. *Appl Environ Microbiol*. 2006;72(7):5069–5072.
- Wang Q, Garrity GM, Tiedje JM, Cole JR. Naive Bayesian classifier for rapid assignment of rRNA sequences into the new bacterial taxonomy. *Appl Environ Microbiol*. 2007;73(16):5261–5267.
- Atarashi K, Tanoue T, Oshima K, et al. Treg induction by a rationally selected mixture of Clostridia strains from the human microbiota. *Nature*. 2013;500(7461):232–236.
- Narushima S, Sugiura Y, Oshima K, et al. Characterization of the 17 strains of regulatory T cell-inducing human-derived Clostridia. *Gut Microbes*. 2014;5(3):333–339.
- Vital M, Howe AC, Tiedje JM. Revealing the bacterial butyrate synthesis pathways by analyzing (meta)genomic data. *mBio*. 2014;5(2):e00889.
- Jodele S, Zhang K, Zou F, et al. The genetic fingerprint of susceptibility for transplant-associated thrombotic microangiopathy. *Blood*. 2016;127(8):989–996.
- Trapnell C, Pachter L, Salzberg SL. TopHat: discovering splice junctions with RNA-Seq. *Bioinformatics*. 2009;25(9):1105–1111.
- Edgar R, Domrachev M, Lash AE. Gene expression omnibus: NCBI gene expression and hybridization array data repository. *Nucleic Acids Res*. 2002;30(1):207–210.
- Barrett T, Wilhite SE, Ledoux P, et al. NCBI GEO: archive for functional genomics data sets—update. *Nucleic Acids Res*. 2013;41:D991–D995.
- Sullivan A, Edlund C, Nord CE. Effect of antimicrobial agents on the ecological balance of human microflora. *Lancet Infect Dis*. 2001;1(2):101–114.
- Jernberg C, Lofmark S, Edlund C, Jansson JK. Long-term impacts of antibiotic exposure on the human intestinal microbiota. *Microbiology*. 2010;156(Pt 11):3216–3223.
- Jakobsson HE, Jernberg C, Andersson AF, Sjolund-Karlsson M, Jansson JK, Engstrand L. Short-term antibiotic treatment has differing long-term impacts on the human throat and gut microbiome. *PLoS One*. 2010;5(3):e9836.
- Gao J, Xu K, Liu H, et al. Impact of the gut microbiota on intestinal immunity mediated by tryptophan metabolism. *Front Cell Infect Microbiol*. 2018;8:13.
- Sikander A, Rana SV, Prasad KK. Role of serotonin in gastrointestinal motility and irritable bowel syndrome. *Clin Chim Acta*. 2009;403(1–2):47–55.
- Mawe GM, Hoffman JM. Serotonin signalling in the gut—functions, dysfunctions and therapeutic targets. *Nat Rev Gastroenterol Hepatol*. 2013;10(8):473–486.
- Mathewson ND, Jenq R, Mathew AV, et al. Gut microbiome-derived metabolites modulate intestinal epithelial cell damage and mitigate graft-versus-host disease. *Nat Immunol*. 2016;17(5):505–513.
- Hooper LV, Littman DR, Macpherson AJ. Interactions between the microbiota and the immune system. *Science*. 2012;336(6086):1268–1273.
- Hughes ER, Winter MG, Duerkop BA, et al. Microbial respiration and formate oxidation as metabolic signatures of inflammation-associated dysbiosis. *Cell Host Microbe*. 2017;21(2):208–219.
- Lanis JM, Alexeev EE, Curtis VF, et al. Tryptophan metabolite activation of the aryl hydrocarbon receptor regulates IL-10 receptor expression on intestinal epithelia. *Mucosal Immunol*. 2017;10:1133–1144.
- Vujkovic-Cvijin I, Dunham RM, Iwai S, et al. Dysbiosis of the gut microbiota is associated with HIV disease progression and tryptophan catabolism. *Sci Transl Med*. 2013;5(193). 193ra91.